

Yeeun Choe

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EDUCATION

Yonsei University, Seoul, South Korea Master of Physical Education in Biomechanics <i>Thesis: "Neuromuscular response of the plantar flexor muscle-tendon unit to unanticipated ground-level perturbation during repeated hopping"</i>	<i>Mar 2024 – Aug 2026</i>
CHA University, Pocheon, South Korea Bachelor of Science in Sports Medicine	<i>Mar 2019 – Feb 2023</i>

RESEARCH INTERESTS

Biomechanics, Human Locomotion, Dynamic Stability & Perturbation Response, Neuromuscular Control, Movement Performance, Injury Risk Prediction, Explainable Machine Learning

AWARDS AND HONORS

Master's Student Research Fellowship <i>Humanities and Social Sciences Division (Arts/Physical Education), National Research Foundation of Korea</i>	<i>2025 - 2026</i>
Best Presentation Award <i>2025 KSBB International Conference, Korean Society of Sport Biomechanics</i>	<i>2025</i>
Summit Scholarship <i>Oneness Mission Club</i>	<i>2024 – 2026</i>
Research Assistant Scholarship <i>Department of Physical Education, Yonsei University</i>	<i>2024 – 2025</i>
Best Presentation Award (Co-author) <i>2024 KSBB Fall Conference, Korean Society of Sport Biomechanics</i>	<i>2024</i>
Top Remnant Scholarship <i>Immanuel Seoul Scholarship Committee, World Evangelization Evangelism Alliance</i>	<i>2019, 2022</i>
Student Representative Scholarship <i>Department of Science in Sports Medicine, CHA University</i>	<i>2021 - 2022</i>
Best Presentation Award <i>Department of Science in Sports Medicine, CHA University</i>	<i>2021</i>
Silver Award <i>20th National Undergraduate Kinesiology Conference, Korean Association of Certified Exercise Professionals</i>	<i>2021</i>

WORK & RESEARCH EXPERIENCE

Frontier Research Institute of Convergence Sports Science (FRICSS), Yonsei University Research Assistant Seoul, South Korea	<i>Mar 2024 – Aug 2026</i>
- Supported experimental data collection using EMG, ultrasound imaging, motion capture, and isokinetic dynamometry across multiple biomechanics studies <i>Related publication:</i> Jo, I., Seiberl, W., & Lee, H.-D. (2025). "Modulation of stretch activation influences the stretch-shortening cycle effect in <i>in vivo</i> human knee extensors." <i>Physiological Reports</i>, 13, e70377. <i>Related publication:</i> Mun, S. I., Jo, I., & Lee, H.-D. (2025). "Lower limb muscle activity during standing long jump performance in children with idiopathic toe walking." <i>The Korean Journal of Physical Education</i>, 64(5), 409–419.	
Yonsei Campus Fitness Center Personal Trainer Seoul, South Korea	<i>May 2024 – Aug 2026</i>
- Designed individualized exercise programs for university students and staff, focusing on strength training, Pilates, and functional movement patterns - Provided hands-on coaching and biomechanical feedback during sessions, tailoring exercise protocols to individual needs, goals, and progress	

2024 Yonsei Junior Convergence Research Group | *Researcher* | Seoul, South Korea

May 2024 – Feb 2025

- Selected for competitive interdisciplinary program and collaborated with Sports Psychology Lab and Lifetime Sports Lab on older adults gait analysis (ICONS-funded)
- Enhanced convergence research expertise through ICONS Academy and international conference participation

Jian Rehabilitation Clinic (Bundang H Hospital) | *Exercise Physiologist* | Seongnam, South Korea

Sept 2021 – Jul 2022

- Designed and implemented exercise protocols for TKR (Total Knee Replacement) and THR (Total Hip Replacement) inpatients, guiding progressive rehabilitation programs
- Collaborated with physicians and physical therapists to develop comprehensive rehabilitation plans, optimizing orthopedic patient recovery through interdisciplinary team care

CHA University: Prof. Suk-Yool Jung | *Undergraduate Research Assistant* | Pocheon, South Korea

Apr 2021 – Jun 2021

- Supported EEG data collection to investigate brain function changes in VR-based spatial perception training study (N=20)
- Assisted experiments as participant preparation coordinator, supporting VR intervention protocols and research procedures

CHA University | *Student Worker* | Pocheon, South Korea

Oct 2020 – Aug 2021

- Managed administrative operations for the Sports Medicine Department, coordinating schedules for 200+ students across academic lectures and educational programs
- Maintained and calibrated specialized biomechanical research equipment including EMG, EEG, Foot plantar pressure systems, and VR setups for ongoing studies

TEACHING EXPERIENCE

Writing Mentoring Program, Yonsei University | *Mentor* | Seoul, South Korea

Mar 2026 – Jun 2026

- Delivered 60 hours of one-on-one academic writing mentorship to three graduate students, providing structured feedback on drafts and guiding iterative revision of research proposals and manuscripts

Yonsei University: Prof. Hae-Dong Lee | *Teaching Assistant* | Seoul, South Korea

Mar 2024 – Dec 2025

- Evaluated assignments and exams for Biomechanics I & II courses, providing feedback to 30+ undergraduate students
- Instructed students on motion capture system (Vicon, OpenCap) for movement analysis, demonstrating Excel and MATLAB data processing workflows

CONFERENCE PRESENTATIONS

“Effect of Predictability of Ground Height Perturbations on Neuromechanical Responses of Lower Extremity Joints During Hopping: A Pilot Study”

2025

2025 KSBB International Conference, Korean Society of Sport Biomechanics

“The Impact of Dual-Tasking on Visual Attention and Reaction Time in Older Pedestrians”

2025

The 92nd Conference of Korean Society of Transportation, Korean Society of Transportation

PUBLICATIONS

Han, S., Jo, I., Kim, T., Kim, J., Cho, Y., & Choe, Y. (2025). “Influence of Subjective Health Status on Positive Psychological Capital in Middle-Aged Adults: Mediating Effect of Walking Self-Efficacy Moderated by Exercise Frequency.” *Journal of Wellness*, 20(2), 395–401. <https://doi.org/10.21097/ksw.2025.5.20.2.395>

SKILLS

Instrumentation: Motion Capture (Vicon), EMG (Delsys), Ultrasound Imaging System (Telemed), Eye tracker (Ergoneers), Isokinetic Dynamometry (CON-TREX® MJ), EEG (DSI-24)

Technical: Python, Matlab, R, SPSS, SAS

Software: OpenSim, UltraTimTrack, OpenCap, Mokka, Kinovea, D-Lab

Language: Korean (native), English (intermediate)